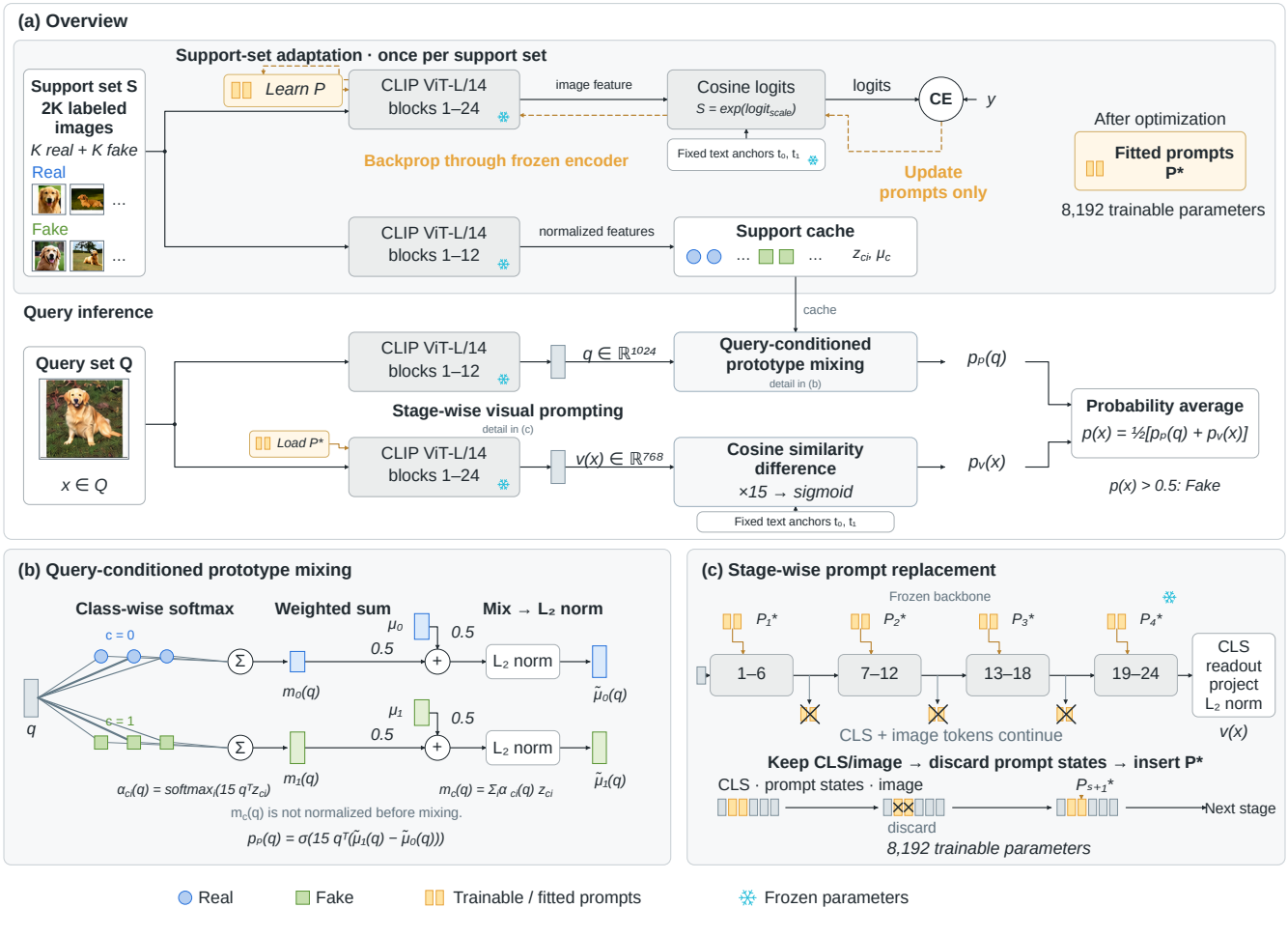

(a) Overview
Support-set adaptation · once per support set
Support set S
2K labeled images
K real + K fake
Real
Fake
Learn P
CLIP ViT-L/14
blocks 1–24
image feature
Cosine logits
S = exp(logit_scale)
logits
CE
y
Fixed text anchors t₀, t₁
Backprop through frozen encoder
Update prompts only
After optimization
Fitted prompts P*
8,192 trainable parameters
CLIP ViT-L/14
blocks 1–12
normalized features
Support cache
z_ci, μ_c
cache
Query inference
Query set Q
x ∈ Q
CLIP ViT-L/14
blocks 1–12
q ∈ ℝ^1024
Query-conditioned prototype mixing
detail in (b)
p_P(q)
Stage-wise visual prompting
detail in (c)
Load P*
CLIP ViT-L/14
blocks 1–24
v(x) ∈ ℝ^768
Cosine similarity difference
×15 → sigmoid
Fixed text anchors t₀, t₁
p_v(x)
Probability average
p(x) = ½[p_P(q) + p_v(x)]
p(x) > 0.5: Fake
(b) Query-conditioned prototype mixing
Class-wise softmax
Weighted sum
Mix → L₂ norm
c = 0
c = 1
q
μ₀
μ₁
0.5
0.5
0.5
0.5
L₂ norm
L₂ norm
m₀(q)
m₁(q)
μ̃₀(q)
μ̃₁(q)
α_ci(q) = softmax_i(15 qᵀz_ci)
m_c(q) = Σ_i α_ci(q) z_ci
m_c(q) is not normalized before mixing.
p_P(q) = σ(15 qᵀ(μ̃₁(q) − μ̃₀(q)))
(c) Stage-wise prompt replacement
Frozen backbone
P₁*
P₂*
P₃*
P₄*
1–6
7–12
13–18
19–24
CLS readout project L₂ norm
v(x)
CLS + image tokens continue
Keep CLS/image → discard prompt states → insert P*
CLS · prompt states · image
discard
P_s+1*
Next stage
8,192 trainable parameters
Real
Fake
Trainable / fitted prompts
Frozen parameters